

as well as wireless remote controls. The controls were extensive with volume, treble and bass controls on both the wired and IR units. The Logitech Z-2300 came with a cool wired control pod that featured a large volume dial and a smaller bass dial.

Two sets offered FM tuners. Both were from Philips—the MMS171FM and the MMS177FM. Surprisingly, all the high-end models excluded this feature. While this is not an essential feature, it is a nice addition that we can only encourage.

Even a 2.1 set needs to stand on its own two feet, and stands for speakers are vital, especially for the subwoofer. The stands on the Altec Lansing FX-6021 impressed us. They were solid metal and of good build quality. A surprise was the high-end MX-5021; this Altec Lansing model lacked any form of satellite stand. To the set's defence, however, we didn't encounter any stability issues while testing. The Logitech X-230 deserves a mention as well: the stand quality was good, and even had a swivel base for wall mounting.

Of Beasts And Beauties

On opening the box marked "Altec Lansing MX-5021," we were struck by the sheer beauty of the set. Even the word "hot" doesn't do justice to describe its dull-finished all-black subwoofer; it

had a sedate cloth grille that barely concealed the silver-coated 6.5-inch driver. Its shiny black satellites had a similar cloth grille wrap, showing off the shiny 1-inch tweeter within, and the dull aluminium-finished 3-inch mid-range drivers (each satellite has two mid-range driv-



ers). The Altec Lansing FX-6021 was the other attention grabber. It was decked in a sheer metallic grey/silver combination. The satellites showed off six tweeters housed within. Another tiny beauty was the Logitech X-230 with a very attractive satellite design in all black.

The lovable beast of the lot was the Logitech Z-2300. This set boasted the largest subwoofer across the entire 2.1 range, at eight inches.

If we had to choose the sleekest, our unanimous choice would be the Creative I-trigue 3400. Its white/grey colours are reminiscent of Apple Macs with uniquely shaped satellites, while its control pod maintains a traditional look.

Performance

The Umax PowerBeats UPB 980 just could not hold its own in our game tests. The sound was tinny and very artificial. To be honest, we didn't expect anything else from this speaker set, considering its low price tag and its sub-standard build quality.

The Altec Lansing MX-5021 was the best performer in this test. Our game tests underlined its solid output—reproducing larger-than-life explosions and lending a solid aural punch to rockets zooming past.

In the DVD movie test, too, the Umax UPB 980 along with the Zebronics ZEB-1400W performed poorly and scored very low points. They could not reproduce the faint footsteps in the *Behind Enemy Lines* movie, nor the roar of the truck in *The Fast and the Furious*. On the other hand, the Altec Lansing MX-5021 and the Logitech Z-2300 performed the best in this test. The sounds produced by these speakers made watching these movies an immersive experience.

How We Tested

Our test PC consisted of a Intel 3.6 GHz Pentium 4 CPU, 1 GB of RAM, a 120 GB Seagate HDD, an ATI X1300 graphics card, and a Creative Audigy 2 Platinum sound card running Windows XP Professional with Service Pack 1.

We put the test candidates through an comprehensive set of intensive tests including a couple of DVD movie tests, multiple music tests and a game test.

The game test consisted of the *Half-life 2* demo—d3_c17_12. We ran this demo with sound settings at "high." The demo contains a lot of multiple sounds, blending voice commands with firing and explosions, and is put to good use to test the capabilities of a speaker.

A DTS sampler disc with a couple of DVD movie clips (from *Behind Enemy Lines* and *The Fast and the Furious*) and a 5.1 soundtrack were used as the DVD test material. This sampler disc contains high quality content and is bundled with high-end Creative sound cards—perfect for stress-testing speakers. The soundtrack was rated under bass, treble and vocals.

For music playback we divided the tests into audio CD and MP3. This was further divided into Hindi and English tracks, and sub-divided into vocal-specific and instrument-specific for English tracks and contemporary and classical for Hindi tracks. The MP3s were encoded at high quality settings at 192 Kbps.

For the English CD we used Enigma's *Sadness 1*, an extremely instrument-intensive track that tests a speakers set's ability to reproduce a number of different musical instruments, and The

Corrs' *Brid Og Ni Mhaille*, which is very vocal-centric Gaelic folk music track. For the Hindi CD we used *Aapki Kashish*, for testing vocals. This track also has some deep drum beats good for bass testing. We chose Ustad Zakir Hussain's *tabla* to test sound reproduction of each note.

English MP3's were *Hotel California* by the Eagles—a good mix of vocal and instruments, especially the drum and guitar accompaniments, and Whitney Houston's *I will always love you*, one of the most stressing vocal tracks, as the singer's voice, though rich, can take extreme high notes. On the Hindi MP3 front, we used *Aashiq banaya aapne* to see how a popular track would sound. This track has some beats and instruments with vocal emphasis. For classical, we used Pandit Bhimsen Joshi's rendition to see whether the entire range of his tone could be accurately reproduced.

We also used special test files of a constant frequency to test low, mid and high frequencies across the band. These were from 50 Hz to 15 KHz.

Last, we stress-tested the bass and treble drivers of all the speakers by playing special THX-certified bass and treble tracks.

The tests were carried out at varying volume levels, and in the case of separate bass and treble controls, these were kept at 50 per cent through the first run and 100 per cent through the second.

We also noted the different features (or lack of them) for the different speakers such as the various inputs, connectivity, controls, accessories, and build quality.